10. GPT & Causal LMs

Overview

Causal language models predict the next token given only past context. GPT-style decoders use triangular attention masks, byte-pair tokenization, and autoregressive sampling (greedy, top-k, nucleus). We contrast pre-training objectives with BERT and set up instruction fine-tuning paths.

Lab: Lab 10.

Prior modules: Module 04, **Module 09**.

Contents

Module track

This module builds on **Module 04**, **Module 09**. Read those PDFs first. References to other modules appear as **Module NN** (not page numbers, because each module is its own PDF).

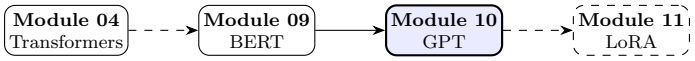


Figure 1: Module sequence (solid box: this PDF; dashed: typical next step).

1 Causal Language Modeling

GPT-2 [4] models $P(x_1, \ldots, x_T) = \prod_{t=1}^{T} P(x_t \mid x_{<t})$ with a **causal** (left-to-right) mask. Contrasts with bidirectional BERT in **Module 09** (masked LM objective).

2 Transformer Decoder Stack

Uses masked multi-head self-attention from **Module 04**:

$$\mathbf{M}_{ij} = \begin{cases} 0 & i \geq j \\ -\infty & i < j \end{cases}, \quad \text{Attn} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} + \mathbf{M}\right)\mathbf{V}. \tag{1}$$

Position i cannot attend to future positions $j > i$.

Note. Causal masking enables autoregressive generation, the same paradigm as RNN decoders in **Module 03** (encoder-decoder setup).

3 Training Objective

Minimize cross-entropy over all non-padding tokens:

$$\mathcal{L} = -\frac{1}{T}\sum_{t=1}^{T} \log P_\theta(x_t \mid x_{<t}). \tag{2}$$

WebText-scale corpora yield emergent few-shot abilities at scale.

4 Generation Strategies

Greedy: $x_t = \arg\max P(x_t \mid x_{<t})$, repetitive. Top-$k$: sample from k highest-probability tokens. Nucleus (top-p): sample from smallest set with cumulative prob $\geq p$:

$$\mathcal{V}^{(p)} = \min\left\{\mathcal{V}' : \sum_{x \in \mathcal{V}'} P(x) \geq p\right\}. \tag{3}$$

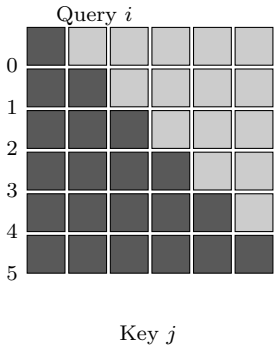
5 BERT vs. GPT

$$\underbrace{P(x_t \mid \mathbf{x}_{\setminus t})}_{\text{BERT MLM}} \quad \text{vs.} \quad \underbrace{P(x_t \mid x_{<t})}_{\text{GPT CLM}}. \tag{4}$$

GPT excels at generation; BERT excels at understanding tasks with full context.

Table 1: Decoding strategies.

Method	Diversity	Coherence
Greedy	Low	Medium
Beam search	Low	High
Top-p	High	Medium to High



White = allowed ($i \geq j$); dark = masked.

Figure 2: Lower-triangular causal attention mask.

6 Fine-Tuning and Instruction Tuning

Supervised fine-tuning (SFT) on labeled pairs; RLHF optional for alignment. Leads to PEFT methods in **Module 11** (LoRA definition) when full fine-tune is costly.

7 Lab

Lab 10 fine-tunes a small GPT on instruction data; experiment with top-p from (4).

8 Scaling Laws Preview

Loss scales as power law with model size N and data D:

$$\mathcal{L}(N, D) \approx \left(\frac{N_c}{N}\right)^{\alpha_N} + \left(\frac{D_c}{D}\right)^{\alpha_D}. \tag{5}$$

Larger GPT models exhibit in-context learning without gradient updates.

8.1 KV Cache

Autoregressive inference caches key/value tensors per layer to avoid recomputing prefixes.

9 Extended Intuition

GPT models are decoder-only Transformers trained to maximize likelihood of the next token given all prior tokens. Causal masking enforces that position i cannot attend to $j > i$, preserving autoregressive semantics at training time. Pre-training on large corpora learns grammar, facts (imperfectly), and style; fine-tuning adapts behavior to instructions or domains.

Generation at inference repeatedly appends the sampled token and runs another forward pass (or uses KV-cache to avoid recomputing past keys/values). Sampling strategies matter: greedy argmax is deterministic but dull; top-k and nucleus (top-p) truncate the tail of the distribution for diversity. Temperature scales logits before softmax: $T > 1$ flattens (more random), $T < 1$ sharpens (more conservative).

Instruction fine-tuning pairs prompts with desired completions; loss is typically computed only on completion tokens, not the prompt. BERT in **Module 09** uses bidirectional context for understanding; GPT trades that for generative coherence. Parameter-efficient updates in **Module 11** matter when full GPT-2/GPT-2-medium fine-tune exceeds VRAM.

9.1 Decoding as Repeated Forward Passes

Autoregressive generation appends one token at a time; complexity without cache is $O(L \cdot T^2 \cdot d)$ per new token at length T. Beam search keeps top-k partial hypotheses; wider beams improve quality on translation-like tasks but hurt diversity on open-ended chat. Repetition penalty down-weights logits of tokens already in context; useful when fine-tune data contains templated phrases. Stop when EOS is sampled or `max_new_tokens` reached; conflating context length with generation length causes silent truncation bugs. Chat templates (post GPT-2 era) wrap user turns with special tokens; fine-tune format must match inference template or the model ignores instructions. RLHF (not in base lab) refines GPT models with human preference rewards; instruction fine-tune here is supervised-only precursor work.

10 Numerical Walkthrough

GPT-2 small: $L = 12$, $d = 768$, vocab $V = 50257$, context $T = 512$, batch $B = 4$ for fine-tune.

Next-token loss on sequence $y_{1:T}$:

$$\mathcal{L} = -\sum_{t=1}^{T-1} \log P_\theta(y_{t+1} \mid y_{1:t}). \tag{6}$$

If model assigns prob 0.01 to true token, contribution is $-\log(0.01) \approx 4.6$ nats per token.

Top-p sampling with $p = 0.9$: sort probabilities descending, keep smallest set whose cumulative mass ≥ 0.9, renormalize, sample. Example logits after softmax: $[0.4, 0.3, 0.15, 0.1, 0.05]$; cumulative $[0.4, 0.7, 0.85, 0.95, \ldots]$; keep first 4 tokens, zero rest.

Fine-tune 1k instruction pairs, $T = 256$, lr $= 5 \times 10^{-5}$, 3 epochs on GPT-2: train loss should fall from ≈ 3.5 toward 1.5 to 2.0 depending on task complexity. KV-cache at inference: store past key/value tensors per layer; each new token adds $O(L \cdot d \cdot T)$ memory but saves $O(T^2)$ recomputation.

Perplexity PPL $= \exp(\bar{\ell})$: if $\bar{\ell} = 2.3$ nats, PPL $\approx e^{2.3} \approx 10$ (model as confused as uniform over 10 tokens). Instruction fine-tune with prompt length 64 and answer length 128: mask prompt labels with `-100` so loss reflects only completion tokens. GPT-2 uses learned positional embeddings up to 1024 positions; fine-tuning at $T = 256$ never updates positions 256..1023 unless you extend context deliberately.

Byte-level BPE merges bytes not Unicode code points; emoji and rare scripts split into multiple ids, inflating sequence length vs character count. Rolling your own training loop: shift logits right by one so position t predicts token $t + 1$; off-by-one misalignment yields loss around $\log V \approx 11$ nats for GPT-2 vocab. Beam search keeps k hypotheses; each step expands $k \times V$ candidates then prunes; use only at inference, never during fine-tune teacher forcing. Contrastive pre-training (not GPT-2) uses bidirectional context; do not confuse OpenAI GPT-2 checkpoints with GPT-3/4 API models when citing capabilities.

11 PyTorch Implementation Notes

- `GPT2LMHeadModel.from_pretrained("gpt2")` with `AutoTokenizer`; set `pad_token = eos_token`.
- Labels for causal LM: same as `input_ids`, shift internally; use `-100` index to ignore prompt tokens in loss.
- Generation: `model.generate(max_new_tokens=50, do_sample=True, top_p=0.9, temperature=0.8)`.
- `past_key_values` returned when `use_cache=True`; pass into next forward for incremental decoding.
- Byte-level BPE means tokenizer quirks on whitespace; always decode with `tokenizer.decode(ids, skip_special_tokens=True)` for inspection.

```
labels = input_ids.clone()
labels[:, :prompt_len] = -100  # mask prompt in loss
out = model(input_ids, labels=labels)
loss = out.loss
```

12 Local GPU Constraints

GPT-2 (124M params) fine-tune at $T = 256$, $B = 2$, fp16 fits 4 GB with gradient accumulation 8. GPT-2 medium (355M) needs LoRA (**Module 11**) or 8-bit loading (`bitsandbytes`) on 1650.

Inference with KV-cache reduces latency; without cache, $O(T^2)$ per generated token becomes painful past $T = 512$. Use `torch.inference_mode()` for generation benchmarks.

13 Debugging Checklist

- Model repeats one phrase: temperature too low with greedy decoding, or insufficient fine-tune data diversity.
- Loss includes prompt tokens: forgot label mask `-100` on prompt positions.
- Garbage unicode output: tokenizer/vocab mismatch between train and generate scripts.
- OOM on long context: reduce T or enable gradient checkpointing in config.
- Fine-tuned model worse than base on general text: catastrophic forgetting; lower lr or fewer epochs.
- Identical completions for different prompts: temperature 0 with greedy decode; raise temperature or enable top-p.

14 Practice Problems

Problem 1. Compute perplexity from average token loss $\bar{\ell} = 2.3$ nats.

Problem 2. Implement top-k ($k = 40$) sampling without `generate()` for one step.

Problem 3. Compare greedy vs top-$p = 0.9$ outputs on 10 prompts after instruction fine-tune.

Problem 4. Measure tokens/sec with and without KV-cache at sequence length 256.

Problem 5. Fine-tune on 500 instruction pairs with loss masked on prompts only vs unmasked full sequence. Compare held-out completion quality.

Problem 6. With `generate()`, sweep temperature $\{0.2, 0.7, 1.0\}$ at fixed top-$p = 0.9$. Tabulate repetition rate on 20 prompts.

Left-padding batched prompts aligns the last token index across rows; right-padded batches misalign causal masks during batched `generate`. Document max context in README when fine-tuning at $T = 256$ on GPT-2 (1024 positions); extrapolation beyond train length degrades coherence. Save generation hyperparameters (`temperature`, `top_p`, `max_new_tokens`) alongside checkpoints for reproducible eval runs. Merge LoRA adapters from **Module 11** before deployment if latency matters; merged GPT-2 weights behave like a standard checkpoint at inference. Log effective tokens/sec with `use_cache=True` vs False at $T = 512$ to quantify KV-cache benefit on your hardware. Cap `max_new_tokens` during eval to prevent runaway generation when EOS logits stay flat after weak fine-tunes. Compare train loss on completion tokens only vs full sequence to verify label mask `-100` is applied correctly.

Run **Lab 10**; compare objectives with BERT in **Module 09**; memory tricks in **Module 11**.

Source certification

This module lists where each core definition, equation, figure, and summary table comes from. Pedagogical simplifications (lab batch sizes, epoch counts, illustrative plots) are labeled in extension sections and are not claimed as optimal production settings.

Table 3: Certified topic map for Module 10.

Topic	Primary source	Where in this PDF
Causal LM / GPT-2 objective	[4]	Eq. 1
Autoregressive decoding	[4]	Section 4
Scaling context (GPT-3)	[1]	Extension section
Contrast with BERT	[2]	Overview

Full bibliographic entries appear below. Figures are schematic unless a caption cites a specific paper figure. If a statement is not listed here, treat it as general ML practice or lab-specific guidance, not a cited theorem.

References

[1] Tom Brown, Benjamin Mann, Nick Ryder, et al. Language models are few-shot learners. In *NeurIPS*, 2020.

[2] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *NAACL*, 2019.

[3] Edward J. Hu, Yelong Shen, Phillip Wallis, et al. LoRA: Low-rank adaptation of large language models. In *ICLR*, 2022.

[4] Alec Radford, Jeffrey Wu, Rewon Child, et al. Language models are unsupervised multitask learners. *OpenAI Technical Report*, 2019.

[5] Ashish Vaswani, Noam Shazeer, Niki Parmar, et al. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.